COURSE (Subject) BOOK

Calculus I

College Name	Education
Department Name	Mathematics
Academic Year	2026 – 2027
Lecturer Name	Ivan Dler Ali and Osman Omer Osman
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Office Hours	Days & Times: 30 Hours
Year of Study	First
Level of Course	Undergraduate

1. Course Information:

Course Title	Calculus I
Course Code	S070401
Credit Hours (ECTS)	6
Semester	1st
Prerequisites	Yes
Theory	4 hours

Practical (if applicable)	**2 hours**

2. Introduction:

This course provides students with the theoretical knowledge and practical skills necessary to understand the fundamental principles and applications of the subject. Through lectures, tutorials, discussions, assignments, exams, and independent study, students will develop analytical thin fundamental principles and applications of calculus. Through lectures, tutorials, discussions, assignments, exams, and independent study, students will develop analytical thinking, problem-solving abilities, and mathematical skills relevant to their academic discipline.

In the first chapter, we study intervals, absolute value, inequalities, functions and their properties, special functions such as trigonometric, logarithmic, and exponential functions, graphs of functions, domain and range, odd and even functions, operations on functions, and inverse functions. These concepts are illustrated through many examples and exercises.

In the second chapter, we study the introduction to limits, the definition of limits, one-sided limits, limits at infinity, the Sandwich Theorem, asymptotes, and continuity. These concepts are illustrated through many examples and exercises.

In the third chapter, we study transcendental functions, including exponential functions, the natural exponential function, logarithmic functions, the natural logarithm function, and hyperbolic functions, as well as some basic concepts of limits. These topics are illustrated through many examples and exercises.

king, problem-solving abilities, and professional competencies relevant to their academic discipline.

3. Aim of the Course:

Calculus is the study of fundamental mathematical concepts related to functions, limits, and continuity. It covers intervals, absolute values, inequalities, the properties and graphs of functions, domains and ranges, even and odd functions, operations on functions, inverse functions, and special functions such as trigonometric, exponential, logarithmic, and hyperbolic functions. It also introduces limits, including limits by definition, one-sided limits, limits at infinity, the Sandwich Theorem, asymptotes, and continuity. The course aims to develop students' understanding of these concepts, strengthen their analytical and problem-solving skills, and enable them to apply calculus techniques through a variety of examples and exercises.

4. Objectives: *(The following is provided as an example only). The objectives are based on the subject specifications.*

Upon completion of this course, students should be able to:

a) Understand intervals, absolute values, inequalities, functions, and their fundamental properties.

b) Determine the domain and range of functions, identify even and odd functions, perform operations on functions, and find inverse functions.

c) Analyze and graph special functions, including trigonometric, exponential, logarithmic, and hyperbolic functions.

d) Evaluate limits using different methods, including the formal definition, one-sided limits, limits at infinity, and the Sandwich Theorem.

e) Identify asymptotes and examine the continuity of functions.

f) Apply mathematical concepts and techniques to solve a wide variety of related problems.

5. Course Learning Outcomes: (The following is provided as an example only). The CLOs are based on the Course specifications.

Category	CLOs	course Learning Outcome
Knowledge	CLO1	**Explain the fundamental concepts of intervals, absolute values, inequalities, functions, limits, continuity, and transcendental functions.**
Intellectual Skills: Application of Intellectual Skills	CLO2	**Apply the properties and operations of functions, limits, and transcendental functions to solve mathematical problems.**
Intellectual Skills: Analysis and Evaluation of Intellectual Skills	CLO3	**Analyze functions by determining their domain, range, symmetry, inverse functions, limits, continuity, and asymptotic behavior.**
Practical Skills	CLO4	**Use appropriate mathematical methods and graphical representations to solve and illustrate problems involving functions and limits.**
Communication skills	CLO5	**Communicate mathematical ideas, solutions, and results clearly using appropriate mathematical language, notation, and graphs.**
Professional/Organizational skills	CLO6	**Demonstrate independent learning, logical thinking, problem-solving skills, and effective time management when completing mathematical exercises and assignments.**

6. Reading and Learning Resources:

Essential Textbooks	• Calculus, HOWARD ANTON, IRL BIVENS, STEPHEN DAVIS.2012, 10th edition. • THOMAS' CALCULUS, Weir Hass, 2005, Pearson Education, Inc. 12th edition. • THOMAS/ FINNEY, George B.Thomas, Ross L. Finney. 1996. 9th edition • Calculus with analytic geometry, George F. Simmons, 1985, by Mc-Grawhill, Inc. • Calculus, Howard Anton, 1995, by Anton text books, Inc.
Recommended Reading	• OpenStax: Calculus, Volume 1, a free textbook covering functions, limits, continuity, derivatives, and related examples. • Paul's Online Math Notes: Calculus I, providing clear explanations, worked examples, practice problems, and solutions related to functions, limits, continuity, and calculus. • MIT OpenCourseWare: Single Variable Calculus, offering free lecture videos, course notes, problem sets, and additional calculus materials. • Khan Academy: Calculus 1, offering instructional videos, exercises, and step-by-step explanations of functions, limits, continuity, and other fundamental calculus topics.
Online Resources	• PowerPoint presentations for course outlines, topics, definitions, graphical illustrations, discussions, and conclusions. The blackboard is used to explain concepts and solve examples. (2 hours of discussion and 4 hours of theoretical instruction).

7. Main Topics of the Subject (course):

Week	Topic
1	Registration
2	Real number and real line, Rules for inequalities, Intervals
3	Solving Inequalities, Absolut value and some properties
4	Definition of Functions, Common function
5	Domain and Rang, Even and odd functions, Operations function, Invers function
6	Limit, Precise definition of a limit, Some basic limit, Sandwich theorem
7	One - sided limits, Infinite Limits, precise definition of Infinite Limits
8	Mid-term Exam
9	Some basic limit of Trigonometric functions
10	Definition of Continuity and some Theorem and examples.

11	Exponential function, natural exponential function,
12	logarithm function, natural logarithm function
13	Hyperbolic function, Some basic of limits
14	Review
15	Finial Exam

8. Assessment Strategy: The breakdown of assessment marks is based on the subject specifications, the University's QA regulations, and the type of study (Bologna, Yearly, or Seasonal).

Assessment	Weight
Discussion	5 Marks
Report and Seminar	5 Marks
Quiz (2 quizzes, 2.5 marks each)	5 Marks
Assignment and Participation	5Marks
Mid-term Examination	20 Marks
Final Examination	60 Marks
Total	**100%**

9. Student Workload Breakdown

Academic Year: 2026-2027		**Semester:** Fall	
ECTS: 6		6*27 = 162 Total number of credit hours	
Prerequisite: Yes			
Type	**No. of Activities**	**Hours (Duration) per Activity**	**Total**
Attendance	**14**	**4 hr**	**4*14=56**
Report	**1**	**10 hr**	**1*10=10**
Mid Term Exam	**1**	**6 hr**	**1*6=6**
Seminar	**1**	**9 hr**	**1*9=9**
Classwork	**6**	**5 hr**	**6*5=30**
Discussion	**14**	**2 hr**	**2*14=28**
Quiz	**3**	**3 hr**	**3*3=9**

Site Visit	**0**	**0**	**0**
Project	**0**	**0**	**0**
Final Exam	**1**	**14 hr**	**1*14=14**
Total			**162hr**

10. Learning Outcome Mapping: Example of Computer Network Course with 6 Learning Outcomes (CLOs)

Topics	Assessment	Week	cLO(s)
Topics	**Assessment**	**Week**	**CLO(s)**
Registration	—	1	—
Real Number and Real Line, Rules for Inequalities, Intervals	Class Participation	2	CLO1, CLO2
Solving Inequalities, Absolute Value and Some Properties	Quiz	3	CLO1, CLO2
Definition of Functions, Common Functions	Discussion	4	CLO1, CLO2, CLO5
Domain and Range, Even and Odd Functions, Operations on Functions, Inverse Functions	Assignment	5	CLO1, CLO2, CLO3, CLO6
Limits, Precise Definition of a Limit, Basic Limits, Sandwich Theorem	Quiz	6	CLO1, CLO2, CLO3
One-Sided Limits, Infinite Limits, Precise Definition of Infinite Limits	Discussion	7	CLO2, CLO3, CLO5
Mid-term Exam	Mid-Term Examination	8	CLO1, CLO2, CLO3
Basic Limits of Trigonometric Functions	Assignment	9	CLO1, CLO2, CLO3,
Definition of Continuity, Theorems and Examples	Quiz	10	CLO1, CLO2, CLO3
Exponential Function, Natural Exponential Function	Discussion	11	CLO1, CLO2, CLO5
Logarithm Function, Natural Logarithm Function	Assignment	12	CLO1, CLO2
Hyperbolic Functions, Basic Limits	Report	13	CLO4, CLO5, CLO6
Review	Class Participation	14	CLO1–CLO6
Final Exam	Final Examination	15	CLO1–CLO6

11. Sample Examination Paper:

Subject: Calculus I First Year Class Time: 60 minutes

Q1/Find the solution set of the following and show their solution set on the real line: (6 marks)

1) $x^2 - 2x + 1 \leq 9$ 2) $\sqrt{(3x-2)^2} = 2 - 3x.$

Q2/ Prove that $\lim_{x\to 5} \sqrt{x-1} = 2$; $\varepsilon = 1$. (4 marks)

Q3/ Let f and g be functions, and let c and L be real numbers. (6 marks)

Is the following prove or disprove? Explain your answer.

a) If $\lim_{x\to c}(f(x) + g(x))$ exists, then $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ must also exist.

b) If $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c}[f(x)]^n = [\lim_{x\to c} f(x)]^n = L^n$, where n is a positive integer.

Q4/ If $f(x)$ is one-to-one, can anything be said about $g(x) = -f(x)$? (4 marks)

Is it also one-to-one? Give reasons for your answer.

Note: This course (subject) book is subject to approval by the Department Council (Scientific Committee) and must be signed by the Head of the Department.